

ST2195 PROGRAMMING FOR DATASCIENCE



**UNIVERSITY
OF LONDON**

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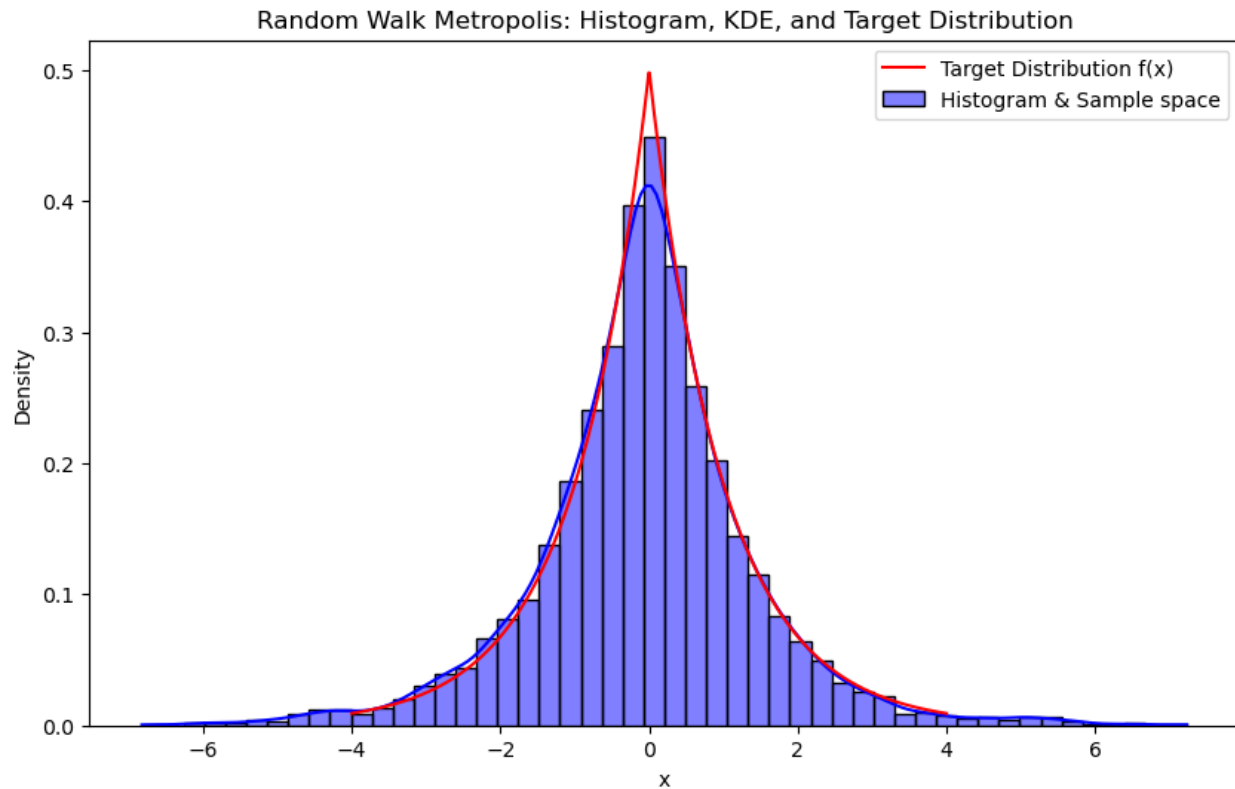
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1. Part 1: Markov Chain Monte Carlo Algorithm

1.1 (a) Random Walk Metropolis Algorithm

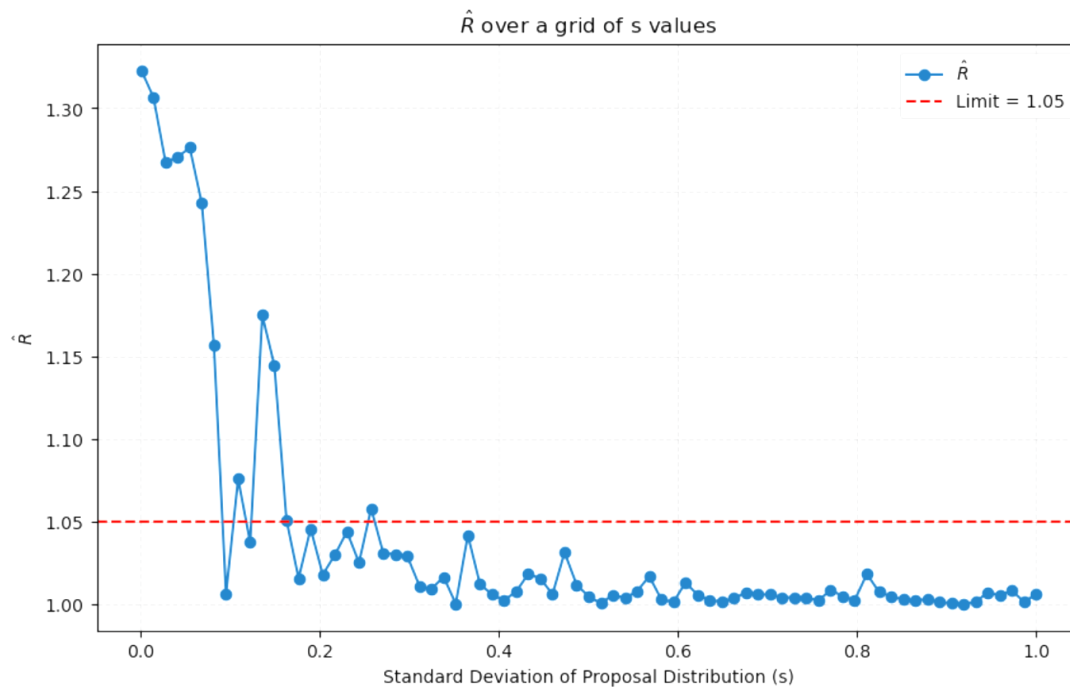
By using the random walk Metropolis algorithm using parameters $N = 10,000$ and $s = 1$, we were able to attain this histogram (blue bars), kernel density plot (blue-line) and an overlay of $f(x)$ (red-line).



The sample mean obtained is 0.013872518379858178 and sample standard deviation obtained is 1.4638411493387042.

1.2(b) $\hat{R}$ Convergence Diagnostic

R hat value obtained when $N = 2000$, $s = 0.001$ and $J = 4$ is 1.006162035961908, which is close to 1, indicating convergence.



Graph above is obtained by plotting the values of $\hat{R}$ over a grid of s values in the interval between 0.001 and 1. The plot does converge towards 1 approximately and it is below 1.05 as shown in the reference line.

2. Part 2: USA Flight Data Analysis

2.1 Introduction

Flight delays are unpleasant, and no one wants to experience it when going on their holidays or business trips. In this report, we aim to analyze the flight data to find the best times and days of the week to minimise delays, evaluate whether older planes suffer more delays on a year-to-year basis, and fit a logistic regression model for the probability of diverted US flights. Through this analysis, we hope to improve the passenger decision-making process.

2.2 (a) What are the best times and days of the week to minimise delays each year?

2.2.1 Cleaning the data

For part 2, we will use the 1995 – 1999 dataset to answer the questions. First, we will read and clean the dataset by cleaning out negative, null and illogical values in our dataset. To answer part a, we will be creating a `delay_df` dataframe that contains the rows 'year', 'month', 'dayofmonth', 'dayofweek', 'deptime', 'arrtime', 'arrdelay'. According to the United States Federal Aviation Administration (FAA) a flight is considered delayed if its arrival time is more than 15 minutes late. Hence, we will be categorizing the delays based on their arrival delays as follows:

- `on_time`: less than 15mins delayed
- `mild_delay`: 15 to 60mins delayed
- `moderate_delay`: 60 to 180mins delayed
- `severe_delay`: more than 180mins delayed

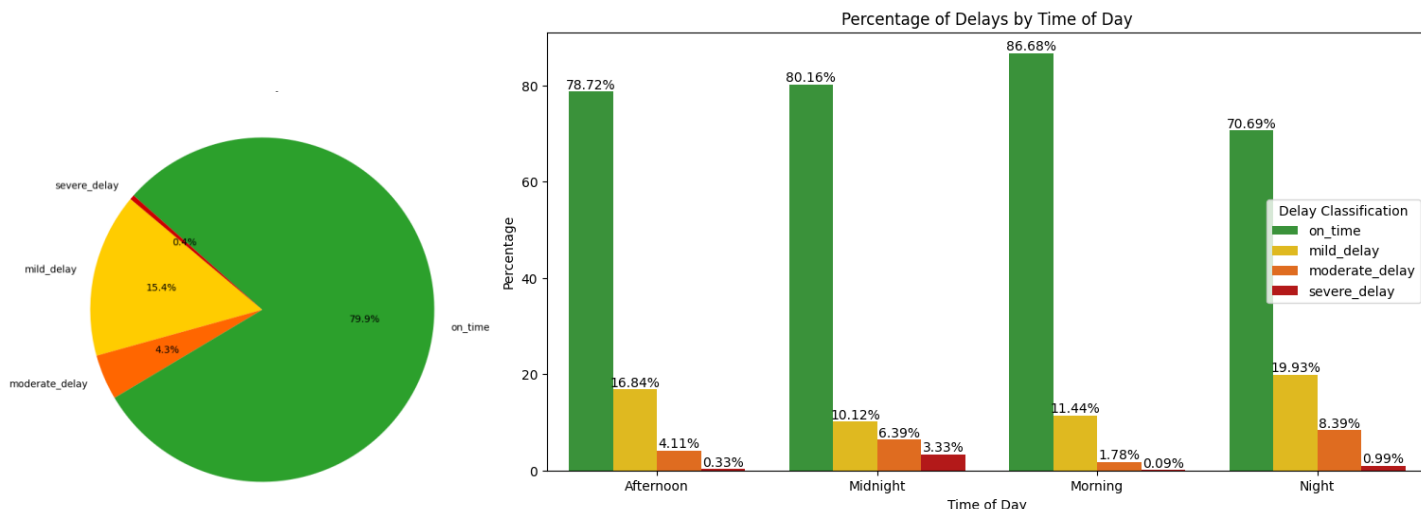
These classifications are then added to a new column 'delay_classification'. To determine the best time to minimise delays, we will be grouping the departure time as follows:

- `Midnight`: 0000 to 0559
- `Morning`: 0600 to 1159
- `Afternoon`: 1200 to 1759
- `Night`: 1800 to 2359

These are then added to a new column "time_of_day". We used departure time as the timing on the tickets are usually the departure time. With these, the dataset is ready to use.

2.2.2 Best time to fly to minimise delays

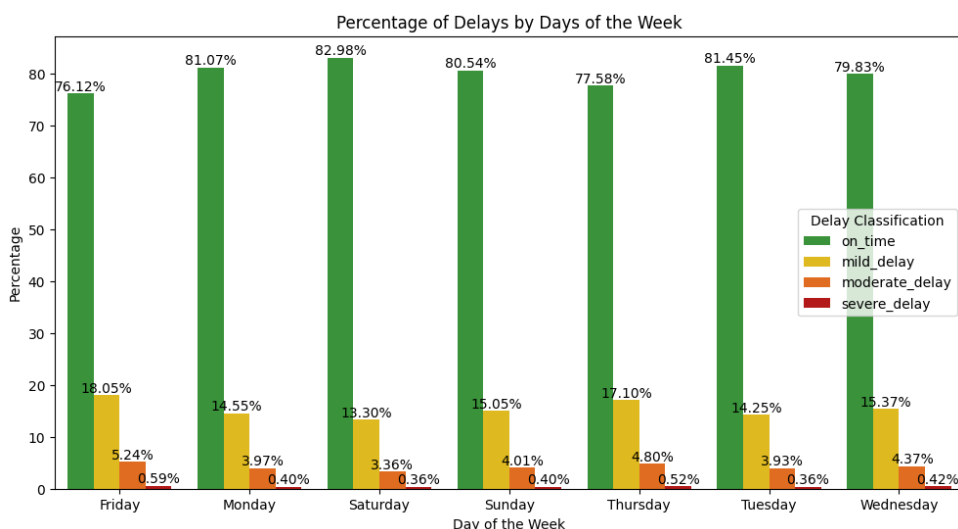
To find the best time to fly with minimal delays, we group the flight records by their time of day and delay classification, before counting the number of flights in each group. After calculating the percentage distribution, we are able to obtain this pie chart that shows the distribution of the delays in our dataset.



After splitting the dataset by the time of the day, we can obtain this bar chart that shows that morning flights have the highest percentage of on-time arrivals. On the other hand, night flights have the lowest proportion. Based on this analysis, it is recommended to book morning flights to minimise the risk of delays.

2.2.3 Best Day of the week to book a plane ticket

To find the best day of the week to book a plane ticket, we group the flight records by the day of the week and delay classification, before counting the number of flights in each group. By calculating percentage distribution, we generated a bar chart that shows us how delays vary across the days of the week.



In 1995, the best day to fly was Saturday with an average arrival delay of 5.32 minutes.
In 1996, the best day to fly was Saturday with an average arrival delay of 6.86 minutes.
In 1997, the best day to fly was Saturday with an average arrival delay of 5.41 minutes.
In 1998, the best day to fly was Saturday with an average arrival delay of 3.56 minutes.
In 1999, the best day to fly was Saturday with an average arrival delay of 5.34 minutes.

In 1995, the best time of day to fly was Morning with an average arrival delay of 2.70 minutes.
In 1996, the best time of day to fly was Morning with an average arrival delay of 4.32 minutes.
In 1997, the best time of day to fly was Morning with an average arrival delay of 3.04 minutes.
In 1998, the best time of day to fly was Morning with an average arrival delay of 2.23 minutes.
In 1999, the best time of day to fly was Morning with an average arrival delay of 2.18 minutes.

The result shows that Saturday is the best day to fly, having the highest percentage of flights arriving on time. In contrast, Friday has the lowest percentage of flights arriving on time. Based on this analysis, it is recommended to book Saturday flights to minimise the risk of delays. The yearly results show that the best day to fly is Saturday while the best time to fly is mornings.

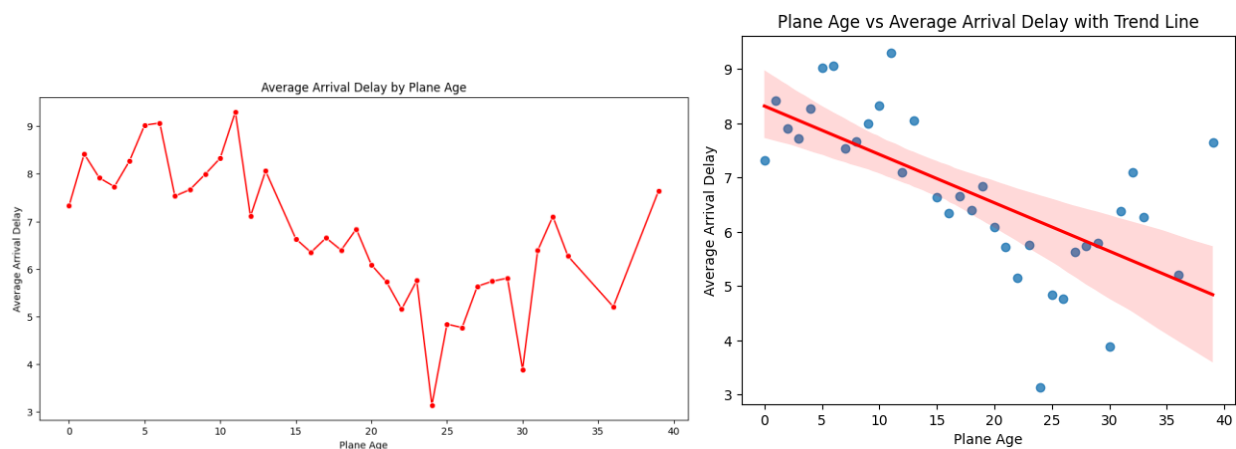
2.3 (b) Evaluate whether older planes suffer more delays on year-to-year basis

2.3.1 Data cleaning

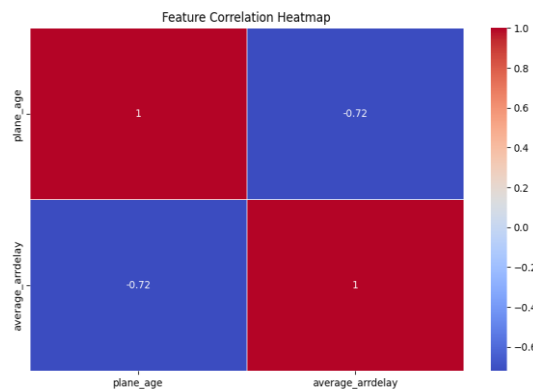
We will be using the airplane's data with the main dataset. After left-joining the airplane's data onto our main dataset and cleaning it, we have a dataset that contains the year of manufacture of the plane, with the original year of the dataset and the delays. Aircraft age is calculated by subtracting the year of manufacture from the year of the dataset. We then grouped the dataset by its tail number to obtain the average delay faced by each aircraft. After removing the ages of planes that are less than 0, the dataset is ready to use.

2.3.2 General trend

By plotting a line chart of average delay against plane age, we are able to see the general trend of how age of airplanes affects delays.

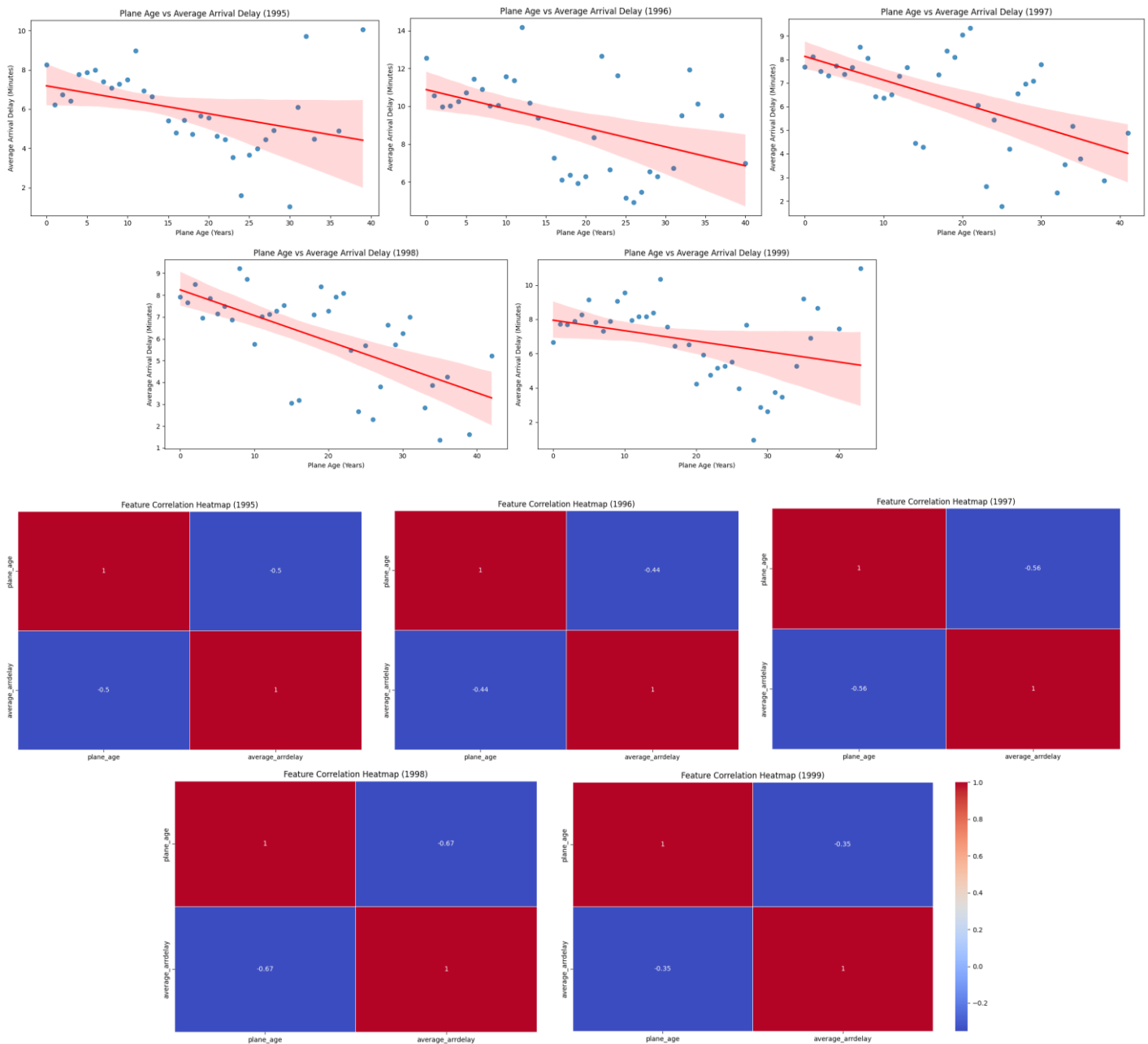


Based on the line graph above, delays tend to decrease as age increases from 0 to 24 years old, indicating that newer planes are more prone to delays. However, after 24 years, delays begin to increase with age. The scatter plot with the regression line supports this observation, showing a downward trend.



To confirm our findings, we plotted a Spearman correlation matrix, as the relationship between plane age and arrival delay is non-linear. This shows a strong negative correlation between plane age and average arrival delays. This indicates that as aircraft age, they generally experience less delays, which supports the trend observed

2.3.3 Throughout the years



These heatmaps throughout the years all support the relationship between age and arrival delay is generally negative. In 1995, 1996, and 1997, the Spearman correlation between plane age and average arrival delay ranges from -0.44 to -0.56, indicating a moderate negative correlation which suggests that older aircraft were associated with fewer delays. In 1998, the correlation strengthens to -0.67, while in 1999, it weakens to -0.35, showing some yearly variation but an overall consistent negative trend.

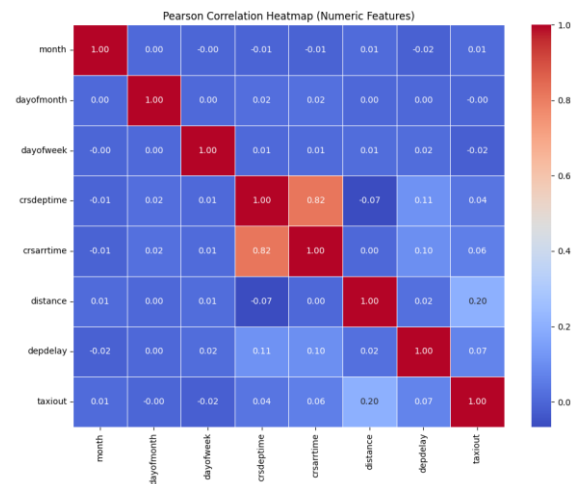
This is quite interesting as most would assume that delays faced would increase as the plane age increases. Some reasons why this may be the case is that newer planes may be assigned to longer and more complicated routes, increasing the chance of delays. Survivorship bias may also play a role as the older planes that are still in service are likely to be more reliable as compared to the newer planes.

2.4 (c) Logistic regression for the probability of diverted US flights

2.4.1 Data cleaning

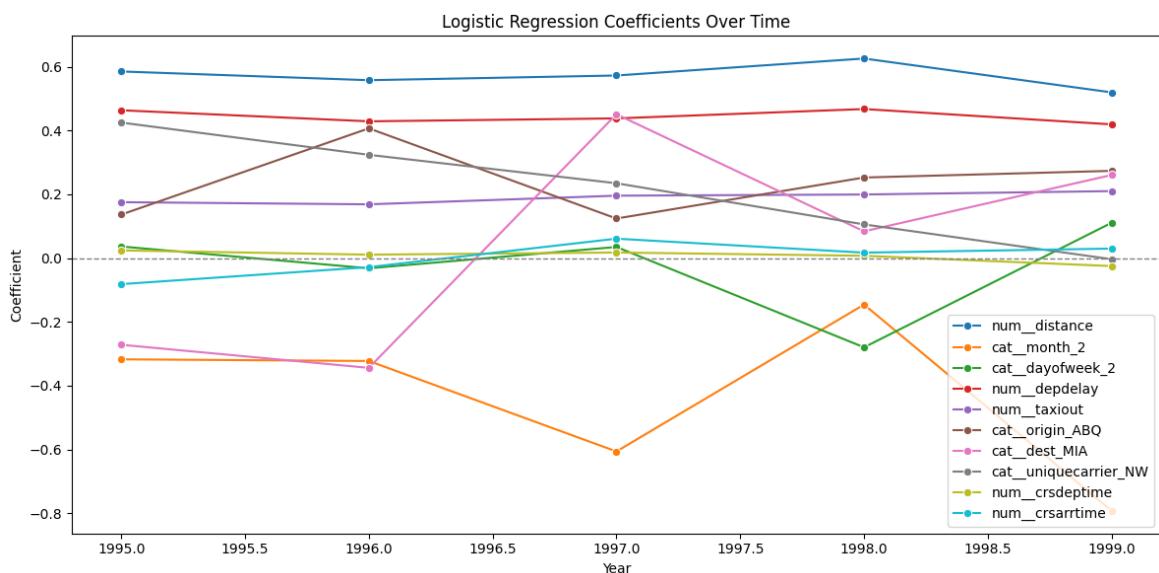
We will be removing cancelled flights as cancelled flights cannot be diverted. After removing null values, our dataset is ready to use.

2.4.2 Features selection



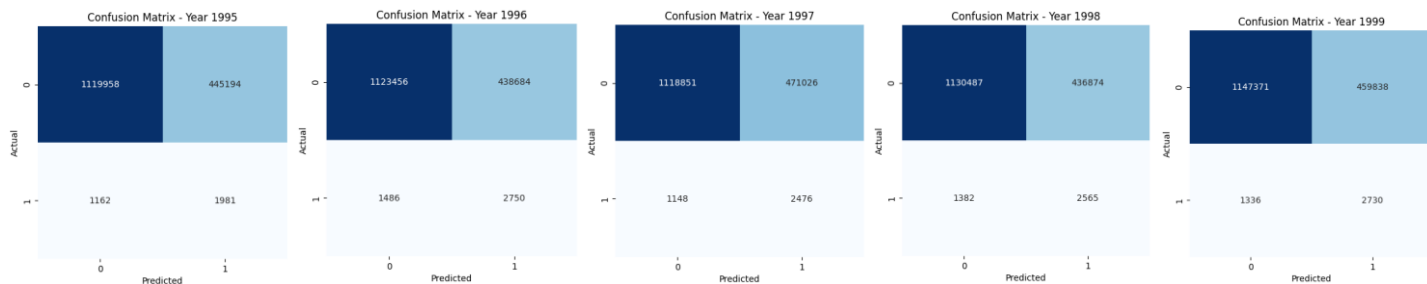
We will be selecting 'month', 'dayofmonth', 'dayofweek', 'crsdeptime', 'crsarrrtime', 'distance', 'uniquecarrier', 'origin', 'dest', 'depdelay', 'taxiout' as our features for our machine learning model. A check for perfect multicollinearity shows that no pairwise correlation equals one, satisfying the assumption of no perfect multicollinearity. However, we will be removing 'dayofmonth' as it has very low correlation and has low predictive value.

2.4.3 Visuals and Scores



The graph shows how the logistic regression coefficient for each feature changed from 1995 to 1999 in predicting flight diversions. 'distance' and 'depdelay' consistently have a strong positive effect, making them reliable predictors. On the other hand, features like 'taxiout' and 'dayofweek' fluctuate between positive and negative, making their influence less stable and reliable for predicting flight diversions. We may consider removing features such as 'crsdeptime' and 'crsarrrtime' for our next model as its coefficient is constantly near 0.

--- Year 1995 ---					--- Year 1996 ---					--- Year 1997 ---					--- Year 1998 ---					--- Year 1999 ---				
	precision	recall	f1-score	support		precision	recall	f1-score	support		precision	recall	f1-score	support		precision	recall	f1-score	support		precision	recall	f1-score	support
Not Diverted	1.00	0.72	0.83	1565152	Not Diverted	1.00	0.72	0.84	1562148	Not Diverted	1.00	0.70	0.83	1589877	Not Diverted	1.00	0.72	0.84	1567361	Not Diverted	1.00	0.71	0.83	1607289
Diverted	0.00	0.63	0.01	3143	Diverted	0.01	0.65	0.01	4236	Diverted	0.01	0.68	0.01	3624	Diverted	0.01	0.65	0.01	3947	Diverted	0.01	0.67	0.01	4866
accuracy			0.72	1568295	accuracy			0.72	1566376	accuracy			0.70	1593581	accuracy			0.72	1571388	accuracy			0.71	1611275
macro avg	0.50	0.67	0.42	1568295	macro avg	0.50	0.68	0.42	1566376	macro avg	0.50	0.69	0.42	1593581	macro avg	0.50	0.69	0.42	1571388	macro avg	0.50	0.69	0.42	1611275
weighted avg	1.00	0.72	0.83	1568295	weighted avg	1.00	0.72	0.83	1566376	weighted avg	1.00	0.70	0.82	1593581	weighted avg	1.00	0.72	0.84	1571388	weighted avg	1.00	0.71	0.83	1611275
Accuracy: 0.7153877299878242					Accuracy: 0.7189882888918453					Accuracy: 0.7036876663397136					Accuracy: 0.7210884180568037					Accuracy: 0.7137831841243736				
F1 Score: 0.0087982270306761					F1 Score: 0.012349969775841112					F1 Score: 0.010378809790286004					F1 Score: 0.01157005408380057					F1 Score: 0.01170081905734014				



These are the results of the logistic regression model. It achieved consistent accuracy around 71% to 72%. However, the accuracy may not be a reliable metric in this case due to the high imbalance in the dataset. The imbalance in dataset also causes the model to perform poorly in identifying diverted flights, with extremely low precision and f1 scores throughout the years. The confusion matrices show strong performance for non-diverted flights but many false negatives for diverted ones. Recall score is about 70% which is good enough as recall is usually low in a highly imbalanced setting. However, it can also not be good enough as the consequence of missing a false negative is very high in our case as the cost of missing a diversion is extremely high.

2.4.4 Further Improvements

As we can see, there is very little diverted compared to non-diverted flights in our dataset, which is causing our precision of our model to be extremely low for predicting diverted flights. Hence, the model can be improved using resampling techniques for imbalanced data, such as SMOTE or random oversampling. SMOTE, which stands for Synthetic Minority Over-sampling Technique, generates fake examples by interpolating between existing minority class samples. Random Oversampling on the other hand duplicates examples from the minority class. Both these examples can help to address the class imbalance and thus, obtain a better precision score.

